

Week 12: Electric Machines

Rotating machine basics, DC motors and generators (shunt, series, separately excited), synchronous machines, induction motors

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 9 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §15.1–15.9

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

By **Vladislav Shkuratov**, made with Claude (AI). Independent study notes, not affiliated with Concordia University or any instructor.

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How to use this deck

- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

Symbols and units

Symbol	Unit	Meaning
n	rpm	speed in revolutions per minute
ω	rad/s	angular speed, $\omega = 2\pi n/60$
T	N·m	torque
P	W or hp	power, $P = T\omega$; 1 hp = 746 W
p		number of poles (always even)
E_a or E_b	V	back EMF (induced armature voltage)
I_a, R_a	A, Ω	armature current and resistance
I_f, R_f	A, Ω	field current and resistance
$k\Phi$	V·s/rad	machine constant times flux: $E_a = k\Phi\omega$, $T = k\Phi I_a$
s		slip of an induction motor

Speed, torque and power

[NOT on formula sheet: memorize]

$$P = T \omega \quad \omega \text{ (rad/s)} = \frac{2\pi n \text{ (rpm)}}{60} \quad 1 \text{ hp} = 746 \text{ W}$$

Trap

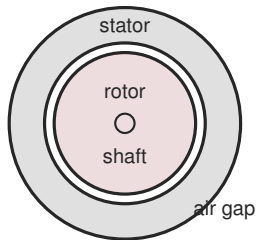
$P = T\omega$ needs ω in **rad/s**. Plugging in rpm is the most common machine-question mistake.

Example 1

A 10 hp motor runs at 1750 rpm. $P = 7460 \text{ W}$, $\omega = 2\pi(1750)/60 = 183.3 \text{ rad/s}$,

$$T = P/\omega = 40.7 \text{ N}\cdot\text{m}$$

Rotating machine basics



Stator: fixed outer part. **Rotor:** spinning part. An air gap separates them.

Field winding: makes the main field.

Armature winding: carries the load current; the back EMF is induced in it.

Synchronous speed with p poles

$$n_s = 120f/p \text{ rpm} \quad (60 \text{ Hz: } 2 \text{ poles } 3600, 4 \text{ poles } 1800, 6 \text{ poles } 1200 \text{ rpm})$$

Where $n_s = 120f/p$ comes from

A 2-pole machine has one north and one south pole. Its magnetic field turns **once per AC cycle**, so it makes f revolutions per second: $60f$ rpm.

With p poles there are $p/2$ north–south pairs around the machine. Each AC cycle moves the field only from one pair to the next, which is $1/(p/2)$ of a turn. So

$$n_s = \frac{60f}{p/2} = \frac{120f}{p} \text{ rpm}$$

Poles p	2	4	6	8
n_s at 60 Hz (rpm)	3600	1800	1200	900

DC machine: the two equations

[NOT on formula sheet: memorize]

Everything comes from these

$$E_a = k\Phi \omega$$

$$T = k\Phi I_a$$

Back EMF is proportional to speed. Torque is proportional to armature current. Same constant $k\Phi$ in both.

Motor (KVL): $V = E_a + I_a R_a$ (like the linear machine, week 11)

Generator (KVL): $V = E_a - I_a R_a$

Engineering shortcut: ratios

Constant flux: $E_{a2}/E_{a1} = n_2/n_1$. Find the new E_a , then scale the speed. No need for $k\Phi$.

Where $E_a = k\Phi\omega$ and $T = k\Phi I_a$ come from

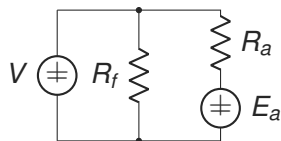
The armature has many conductors cutting the field. From week 11, each conductor gets $e = B\ell u$ (and $u = r\omega$ for a rotor of radius r) and feels $f = Bil$ (torque fr).

Add up all the conductors: B is proportional to the flux Φ , and everything else (number of conductors, ℓ , r , winding layout) is a fixed construction number. Lump it into k :

$$E_a = k\Phi\omega \qquad T = k\Phi I_a$$

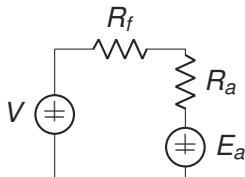
Why the same k : energy conservation. The electrical power converted is $E_a I_a$ and the mechanical power produced is $T\omega$. These must be equal, so $(k\Phi\omega)I_a = (k\Phi I_a)\omega$: the constants must match. So once you find $k\Phi = E_a/\omega$ at one operating point, use it for both speed and torque (Example 2).

DC motor circuit models



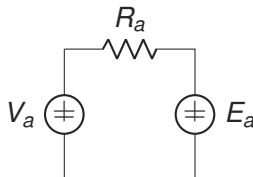
Shunt: field in parallel.

$$I_L = I_f + I_a, \quad I_f = V/R_f$$



Series: field in series.

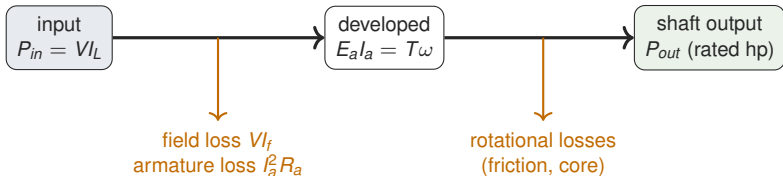
$$I_f = I_a = I_L$$



Separately excited: field on its own supply. Constant flux.

Trap: in a shunt motor the **line** current splits: $I_L = I_f + I_a$. Only I_a flows through R_a and makes torque.

Where the power goes in a DC motor



Electrical side: KVL on the armature, $V = E_a + I_a R_a$. Multiply by I_a : $VI_a = E_a I_a + I_a^2 R_a$. The $E_a I_a$ part is what actually turns into mechanical power.

Mechanical side: the developed torque $T = k\Phi I_a$ is slightly more than the shaft torque; the difference overcomes friction.

That's why, in Example 2, the developed torque (100 N·m) is a bit bigger than the shaft torque (95 N·m).

Try it: Example 2, DC shunt motor

A DC shunt motor is rated 240 V, 1500 rpm, 20 hp, with full-load efficiency 88%. $R_f = 120 \Omega$, $R_a = 0.15 \Omega$.

(a) Line current at full load. (b) Back EMF at rated speed. (c) Developed torque.

(d) The load drops until $I_a = 30 \text{ A}$ (flux constant). New speed?

your work here ↓

Example 2: solution (1/2)

P_{in} rated 20 hp is the **output**:

$$P_{in} = \frac{20 \times 746}{0.88} = \frac{14\,920}{0.88} = 16\,955 \text{ W}$$

(a) input power = VI_L : $I_L = 16\,955/240 = 70.6 \text{ A}$

Split field: $I_f = V/R_f = 240/120 = 2 \text{ A}$; armature gets the rest:
 $I_a = 68.6 \text{ A}$

(b) armature KVL:

$$E_a = V - I_a R_a = 240 - 68.6(0.15) = 240 - 10.3 = 229.7 \text{ V}$$

(c) $k\Phi$ $\omega = 2\pi(1500)/60 = 157.1 \text{ rad/s}$:

$$k\Phi = E_a/\omega = 229.7/157.1 = 1.462$$

Example 2: solution (2/2)

(c) T $T = k\Phi I_a = 1.462 \times 68.6 = 100.4 \text{ N}\cdot\text{m}$

(d) E_a new armature current, same V and R_a :
 $E_{a2} = 240 - 30(0.15) = 235.5 \text{ V}$

Ratio flux constant, so speed is proportional to E_a :
 $n_2 = 1500 \times \frac{235.5}{229.7} = 1538 \text{ rpm}$

Why the speed barely changes

$I_a R_a$ is only a few volts out of 240, so E_a (and the speed) hardly moves. Shunt motors have nearly constant speed from no load to full load.

The developed torque (100.4 N·m) is a bit more than the shaft torque (14 920/157.1 = 95 N·m): the difference covers rotational losses.

Try it: Example 3, separately excited motor

$V_a = 200 \text{ V}$, $R_a = 0.5 \text{ } \Omega$, $k\Phi = 1.8 \text{ V}\cdot\text{s/rad}$ (constant field).

(a) No-load speed (rad/s). (b) Speed when delivering 36 N·m.

your work here ↓

Example 3: solution

(a) no load: almost no torque needed, so $I_a \approx 0$ and $E_a = V_a$:

$$\omega_0 = \frac{V_a}{k\Phi} = \frac{200}{1.8} = 111.1 \text{ rad/s}$$

(b) I_a the torque sets the current: $I_a = T/(k\Phi) = 36/1.8 = 20 \text{ A}$

(b) E_a armature KVL: $200 - 20(0.5) = 190 \text{ V}$

(b) ω $\omega = E_a/(k\Phi) = 190/1.8 = 105.6 \text{ rad/s}$

Series motors

Behaviour

$I_f = I_a$, so the flux grows with the current and $T \propto I_a^2$ (before saturation).

Huge starting torque: ideal for traction (trains, cranes, starter motors).

Speed rises as load drops: with no load, speed can climb dangerously.

Trap

Never run a series motor with no mechanical load: it can “run away” and destroy itself.

Separately excited and shunt motors: speed control by changing V_a (armature voltage) or by weakening the field (less flux gives **higher** speed, since $\omega = E_a/k\Phi$).

Why series motors behave that way

In a series motor the field winding carries the armature current, so the flux is proportional to it (before saturation): $\Phi \propto I_a$.

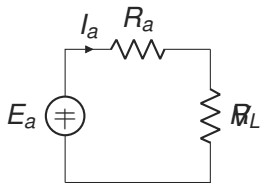
Torque: $T = k\Phi I_a \propto I_a \cdot I_a = I_a^2$. Double the current, four times the torque. At start the current is large, so the torque is huge.

Speed: $\omega = \frac{E_a}{k\Phi}$. With a light load, I_a is small, so Φ is small, and dividing by a small Φ gives a very large speed. With no load at all, the speed keeps climbing: “runaway”.

Shunt, by contrast: the field has its own fixed current V/R_f , so Φ stays constant and the speed barely changes with load.

DC generators

[NOT on formula sheet: memorize]



A prime mover spins the shaft. $E_a = k\Phi\omega$ drives current into the load: $V = E_a - I_a R_a$. The terminal voltage **drops** as load current rises.

Example 4

$E_a = 250$ V at rated speed, $R_a = 0.1$ Ω , load current 100 A: $V = 250 - 10 = 240$ V.

At 80% of rated speed (same flux): $E_a = 0.8 \times 250 = 200$ V.

Synchronous machines (alternators)

[NOT on formula sheet: memorize]

The rotor carries a DC field and spins. Its rotating field induces three-phase AC in the stator.

$$f = p n / 120 \quad n_s = 120 f / p$$

Example 5

For 60 Hz: a 2-pole steam turbine alternator spins at **3600 rpm**; a 72-pole hydro generator at only **100 rpm**. Slow turbines need many poles.

Synchronous motor

Same machine as a motor: runs at exactly n_s regardless of load (up to a limit). Over-excited, it supplies reactive power and can correct power factor.

Induction motors

[NOT on formula sheet: memorize]

The stator's three-phase currents make a field rotating at n_s . Currents are **induced** in the rotor (no connections), and it chases the field, always slightly slower.

$$s = (n_s - n)/n_s \quad \text{rotor frequency} = sf$$

Power flow

Air-gap power P_{ag} crosses to the rotor. Rotor copper loss = $s P_{ag}$. Mechanical power = $(1 - s) P_{ag}$.

Trap: at $n = n_s$ ($s = 0$) the rotor sees no changing field: no current, no torque. An induction motor can never reach n_s .

Why the rotor loses $s P_{ag}$

The same torque T acts across the air gap on both sides:

Air-gap power (field turns at ω_s) $P_{ag} = T \omega_s$

Mechanical power (rotor turns at ω) $P_{mech} = T \omega = T \omega_s(1 - s)$

The difference is lost in the rotor $P_{ag} - P_{mech} = s P_{ag}$

Physically: the rotor slips behind the field at $s n_s$. Its induced currents run at the slip frequency sf and heat the rotor bars. The more slip, the more of the air-gap power is wasted as rotor heat.

That's why induction motors are designed to run at small slip (2–6%): at $s = 0.03$, 97% of the air-gap power becomes mechanical.

Try it: Example 6, induction motor

A 4-pole, 60 Hz induction motor runs at 1746 rpm with 10 kW crossing the air gap.

Find n_s , the slip, the rotor frequency, the rotor copper loss and the mechanical power.

your work here ↓

Example 6: solution

$$n_s \quad 120f/p = 120(60)/4 = 1800 \text{ rpm}$$

Slip how far behind the field, as a fraction of n_s :

$$(1800 - 1746)/1800 = 0.03 \quad (3\%)$$

Rotor f the rotor sees the field pass at the slip speed:

$$sf = 0.03 \times 60 = 1.8 \text{ Hz}$$

Rotor loss $sP_{ag} = 0.03 \times 10 = 0.3 \text{ kW}$

Mechanical $(1 - s)P_{ag} = 0.97 \times 10 = 9.7 \text{ kW}$

Choosing a motor (quick guide)

Motor	Best for
Induction	fans, pumps, compressors, most industrial loads; nearly constant speed
Synchronous	exact constant speed, power factor correction, very large drives
DC shunt / sep. excited	easy wide-range speed control
DC series	very high starting torque (traction, starters)
Brushless DC, stepper	electronics, robotics, positioning (§16)

Motor drives

Modern drives use power electronics to vary the voltage (DC motors) or the frequency (AC motors, since $n_s \propto f$), giving efficient speed control.

Try it: Example 7, efficiency of a motor

A motor draws 50 kW and delivers 45 kW at the shaft at 1200 rpm.

Find the efficiency, the losses, the output in hp, and the shaft torque.

your work here ↓

Example 7: solution

$$\eta \quad P_{out}/P_{in} = 45/50 = 90\%$$

$$\text{Losses} \quad 50 - 45 = 5 \text{ kW}$$

$$\text{hp} \quad 45\,000/746 = 60.3 \text{ hp}$$

$$\text{Torque} \quad P = T\omega \text{ with } \omega \text{ in rad/s: } \omega = 2\pi(1200)/60 = 125.7, \\ T = 45\,000/125.7 = 358 \text{ N}\cdot\text{m}$$

Try it: Example 8, which machine?

For each, say which machine type fits and why:

(a) A generator in a coal plant producing 60 Hz. (b) An electric train. (c) A water pump in a building. (d) A conveyor needing easy speed changes.

your work here ↓

Example 8: solution

- (a) synchronous generator (fixed f from fixed speed)
- (b) series DC (or a modern AC drive): huge starting torque
- (c) induction motor: cheap, rugged, constant speed
- (d) DC shunt / separately excited (or an induction motor with a variable-frequency drive)

Permanent-magnet motors and regulation

[NOT on formula sheet: memorize]

Permanent-magnet (PM) DC motor

Magnets replace the field winding: the flux is fixed, so it behaves exactly like a separately excited motor (constant $k\Phi$). Common in small motors (fans, toys, cordless tools).

Regulation

Speed regulation = $(n_{nl} - n_{fl}) / n_{fl} \times 100\%$ (motors)

Voltage regulation = $(V_{nl} - V_{fl}) / V_{fl} \times 100\%$ (generators)

nl = no load, fl = full load.

Motor dynamics

Newton's law for rotation

$$J(d\omega/dt) = T_{motor} - T_{load} - b\omega$$

J = moment of inertia ($\text{kg}\cdot\text{m}^2$), b = friction coefficient. At steady speed $d\omega/dt = 0$: the motor torque exactly balances load plus friction.

If the load torque suddenly increases, $d\omega/dt < 0$: the motor slows, E_a drops, I_a rises, and torque rises until a new balance is reached. This is the self-regulating behaviour from week 11.

Try it: Example 9, speed regulation

A shunt motor runs at 1575 rpm with no load and 1500 rpm at full load. A DC generator gives 250 V at no load and 240 V at full load.

Find the motor's speed regulation and the generator's voltage regulation.

your work here ↓

Example 9: solution

$$\text{Motor speed regulation} = \frac{n_{nl} - n_{fl}}{n_{fl}} = \frac{1575 - 1500}{1500} = 5\%$$

$$\text{Generator voltage regulation} = \frac{V_{nl} - V_{fl}}{V_{fl}} = \frac{250 - 240}{240} = 4.2\%$$

Both divide by the **full-load** value. Small regulation means the speed (or voltage) barely changes with load; shunt motors typically have 2–8%.

Traps

Check these before you box an answer

- ω in rad/s for $P = T\omega$ and $E_a = k\Phi\omega$.
- Rated power = **output**. Input = P_{rated}/η .
- Shunt: $I_L = I_f + I_a$; only I_a goes through R_a .
- Motor: $V = E_a + I_a R_a$. Generator: $V = E_a - I_a R_a$.
- Constant flux: $E_a \propto \text{speed}$. Use ratios.
- Poles, not pole pairs: $n_s = 120f/p$.
- Slip is a fraction of n_s , not of n .

Exam-format traps

Draw the equivalent circuit

Draw the DC machine circuit with I_L , I_f , I_a labeled. It prevents the most common mistake (using I_L in R_a).

Units

State rpm or rad/s on every speed. Give hp and W when the question mixes them.

Summary

Mechanics	$P = T\omega$, $\omega = 2\pi n/60$, $1 \text{ hp} = 746 \text{ W}$, $\eta = P_{out}/P_{in}$
DC machine	$E_a = k\Phi\omega$, $T = k\Phi I_a$; motor $V = E_a + I_a R_a$, generator $V = E_a - I_a R_a$
Shunt	$I_L = I_f + I_a$, $I_f = V/R_f$; nearly constant speed
Series	$T \propto I_a^2$, high starting torque; never run unloaded
Synchronous	$n_s = 120f/p$; runs exactly at n_s
Induction	$s = (n_s - n)/n_s$; rotor loss sP_{ag} , mechanical $(1 - s)P_{ag}$

Final check

8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

P1: torque

A 5 hp motor runs at 1450 rpm. Shaft torque?

your work here ↓

P2: slip

A 4-pole, 50 Hz induction motor runs at 1440 rpm. Synchronous speed and slip?

your work here ↓

P3: DC shunt motor (exam style)

Rated 500 V, 1800 rpm, 50 hp, efficiency 90%, $R_f = 250 \Omega$, $R_a = 0.1 \Omega$.

(a) I_L (b) I_a (c) E_a (d) the speed when I_a drops to 40 A (constant flux)

your work here ↓

P4: separately excited

$V_a = 240 \text{ V}$, $R_a = 0.4 \text{ } \Omega$, $k\Phi = 2.0$. At $I_a = 25 \text{ A}$: find the torque and the speed (rad/s). What is the no-load speed?

your work here ↓

P5: DC generator

$E_a = 120 \text{ V}$, $R_a = 0.2 \text{ } \Omega$, load current 50 A. Terminal voltage?

your work here ↓

P6: induction speed

A 6-pole, 60 Hz induction motor has 5% slip. Speed and rotor frequency?

your work here ↓

P7: efficiency

A 50 hp motor (37.3 kW out) draws 42 kW. Efficiency?

your work here ↓

P8: alternator

A 6-pole alternator spins at 1000 rpm. What frequency does it produce?

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution

Power $5 \times 746 = 3730 \text{ W}$

Speed rpm to rad/s: $\omega = 2\pi(1450)/60 = 151.8 \text{ rad/s}$

Torque $T = P/\omega = 3730/151.8 = 24.6 \text{ N}\cdot\text{m}$

P2: solution

$$n_s \quad 120(50)/4 = 1500 \text{ rpm}$$

$$\text{Slip} \quad (n_s - n)/n_s = (1500 - 1440)/1500 = 0.04 \text{ (4\%)}$$

P3: solution

(a) input = rated output / η : $\frac{50 \times 746}{0.9} = 41\,444\text{ W}$,

$$I_L = 41\,444 / 500 = \boxed{82.9\text{ A}}$$

(b) field: $I_f = 500 / 250 = 2\text{ A}$, so $I_a = 82.9 - 2 = \boxed{80.9\text{ A}}$

(c) armature KVL: $E_a = 500 - 80.9(0.1) = \boxed{491.9\text{ V}}$

(d) new $E_{a2} = 500 - 40(0.1) = 496\text{ V}$; constant flux:

$$n_2 = 1800 \times \frac{496}{491.9} = \boxed{1815\text{ rpm}}$$

P4: solution

Torque $T = k\Phi I_a = 2.0 \times 25 = 50 \text{ N}\cdot\text{m}$

E_a KVL: $240 - 25(0.4) = 230 \text{ V}$

Speed $\omega = E_a / (k\Phi) = 230 / 2 = 115 \text{ rad/s}$

No load $I_a \approx 0$: $\omega_0 = 240 / 2 = 120 \text{ rad/s}$

P5: solution

KVL generator: the internal drop **subtracts**: $V = E_a - I_a R_a$

Result $120 - 50(0.2) = 110 \text{ V}$

P6: solution

$$n_s \quad 120(60)/6 = 1200 \text{ rpm}$$

$$\text{Speed} \quad n = (1 - s)n_s = 0.95 \times 1200 = 1140 \text{ rpm}$$

$$\text{Rotor } f \quad sf = 0.05 \times 60 = 3 \text{ Hz}$$

P7: solution

$$\eta \quad P_{out}/P_{in} = 37.3/42 = \boxed{88.8\%} \quad (\text{losses: 4.7 kW})$$

P8: solution

Formula $f = \frac{pn}{120}$ (the same relation as $n_s = 120f/p$, solved for f)

Result $\frac{6 \times 1000}{120} =$ 50 Hz